

INFORME DE SEGURIDAD DE RED

Proyecto: Evaluación de Tráfico y Detección de Anomalías

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Fecha: 26/04/2026

1. Resumen Ejecutivo

Durante una sesión de monitoreo de red de **457 segundos**, se identificaron actividades compatibles con **reconocimiento de red** y **comunicación tipo Command & Control (C2) disfrazada en puerto 443 con jitter**.

El patrón observado (bajo volumen, persistente, con intervalos variables de 5–15 segundos hacia un mismo destino) **no corresponde a navegación HTTPS normal** y sugiere un canal encubierto de control.

Riesgo: Alto. Existe probabilidad de compromiso si el comportamiento persiste en producción.

2. Alcance

- **Host analizado:** 192.168.1.88
- **Duración de captura:** 457 s
- **Entorno:** Red local (WiFi)
- **Herramientas:** Wireshark, Nmap
- **Interfaz de captura:** [WiFi / lo / eth0]

3. Metodología

1. Captura de tráfico en tiempo real con Wireshark.
2. Aplicación de filtros:
 - `tcp.flags.syn == 1 && tcp.flags.ack == 0`
 - `tcp.port == 443`
 - `dns`
3. Correlación de eventos por **IP origen/destino, puerto, frecuencia y patrón temporal**.
4. Validación de comportamiento vs. baseline de tráfico normal (navegación web).

4. Hallazgos

4.1 Reconocimiento (Port Scanning)

- Se detectaron paquetes **SYN** dirigidos a múltiples puertos.
- Comportamiento consistente con **escaneo de puertos (Nmap)** previo a intrusión.

Evidencia:

- Secuencias SYN sin establecimiento completo de sesión.
- Múltiples destinos de puerto en un corto periodo.

4.2 Comunicación C2 Disfrazada en 443 (CRÍTICO)

- Tráfico hacia **puerto 443** con:
 - **Intervalos variables:** 5–15 s (jitter)
 - **Bajo volumen**
 - **Persistencia en el tiempo**
 - **Mismo destino** repetido

Interpretación:

Patrón de **beaconing** (llamadas periódicas a servidor de control) diseñado para evadir detección basada en intervalos fijos.

Evidencia:

- Conexiones TCP repetitivas con secuencia handshake + envío breve + cierre.
- Regularidad aproximada (no exacta) dentro de un rango temporal.

5. Análisis

El tráfico en 443 **simula actividad HTTPS**, pero presenta anomalías:

- No depende de interacción del usuario
- No hay diversidad de destinos típica de navegación
- Mantiene **periodicidad aproximada** (característica de C2 con jitter)

La correlación con actividad previa de escaneo sugiere una cadena:

Reconocimiento → Establecimiento de canal C2

No se confirmó explotación de credenciales en este puerto (SSH ≠ 443).

6. Clasificación

- **Tipo de amenaza:** Command & Control (C2) con evasión (jitter)
- **Fase del ataque:** Post-intrusión / comunicación
- **Severidad:** Alta

7. Impacto Potencial

- Recepción remota de comandos
- Exfiltración de datos (bajo volumen)
- Persistencia del atacante en el sistema

8. Recomendaciones

8.1 Contención inmediata

- Bloquear IP/dominio destino identificado
- Restringir salidas no esenciales en puerto 443 (egress filtering)

8.2 Erradicación

- Revisar procesos activos en el host afectado
- Eliminar software sospechoso / no autorizado
- Validar integridad del sistema

8.3 Fortalecimiento

- Implementar monitoreo basado en **comportamiento** (no solo puertos)
- Alertas por **persistencia + baja entropía de destinos**
- Registro y correlación en SIEM (si disponible)

8.4 Credenciales

- Rotación de contraseñas
- Uso de MFA donde aplique

9. Evidencia Adjunta

No.	Time	Source	Destination	Protocol	Length	Info
6213	233.28099	192.168.1.88	192.168.1.70	TCP	66	[TCP Retransmission] 443 → 49408 [ACK] Seq=1 Ack=1 Win=255 Len=1
6221	239.29187	192.168.1.88	192.168.1.70	TCP	66	[TCP Retransmission] 443 → 49408 [ACK] Seq=1 Ack=2 Win=8192 Len=0 SLE=1 SRE=2
6229	247.29213	192.168.1.88	192.168.1.70	TCP	66	[TCP Retransmission] 443 → 49408 [ACK] Seq=875 Ack=8809 Win=65280 Len=0
6243	253.46532	2806:106e:2d:a245::	2001:4c28:3000::	TCP	86	61635 → 443 [FIN, ACK] Seq=8840 Ack=876 Win=64128 Len=0
6246	253.62199	2806:106e:2d:a245::	2001:4c28:3000::	TCP	86	57118 → 443 [TCP Previous segment not captured] 443 → 56487 [FIN, ACK] Seq=8809 Ack=676 Win=64128 Len=31
6247	253.76667	192.168.1.88	107.167.110.216	TCP	66	58066 → 443 [ACK] Seq=876 Ack=8809 Win=65280 Len=0
6252	253.92441	192.168.1.88	107.167.110.216	TCP	66	57612 → 443 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0
6514	355.25875	192.168.1.88	192.168.1.70	TCP	66	57613 → 7680 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0
6518	355.25922	192.168.1.88	192.168.1.70	TCP	66	[TCP Retransmission] 443 → 49408 [ACK] Seq=1 Ack=1 Win=255 Len=1
6528	359.25960	192.168.1.88	192.168.1.70	TCP	66	[TCP Retransmission] 443 → 49408 [ACK] Seq=1 Ack=2 Win=8192 Len=0 SLE=1 SRE=2
6533	362.25983	192.168.1.88	192.168.1.70	TCP	66	[TCP Retransmission] 443 → 49408 [ACK] Seq=875 Ack=8809 Win=65280 Len=0
6545	370.25968	192.168.1.88	192.168.1.70	TCP	66	[TCP Retransmission] 443 → 49408 [ACK] Seq=875 Ack=8809 Win=65280 Len=0
6900	487.47785	192.168.1.88	20.190.155.132	TCP	66	57614 → 443 [FIN, ACK] Seq=8840 Ack=876 Win=64128 Len=0
6927	487.97986	192.168.1.88	20.190.155.132	TCP	66	57615 → 443 [TCP Previous segment not captured] 443 → 56487 [FIN, ACK] Seq=8809 Ack=676 Win=64128 Len=31
6959	489.47227	192.168.1.88	20.3.0.251	TCP	66	57616 → 443 [ACK] Seq=876 Ack=8809 Win=65280 Len=0
7083	551.47381	2806:106e:2d:a245::	2001:4c28:3000::	TCP	86	55390 → 443 [FIN, ACK] Seq=8840 Ack=876 Win=64128 Len=0
7084	551.57621	2806:106e:2d:a245::	2001:4c28:3000::	TCP	86	64420 → 443 [TCP Previous segment not captured] 443 → 56487 [FIN, ACK] Seq=8809 Ack=676 Win=64128 Len=31
7085	551.77598	192.168.1.88	107.167.110.216	TCP	66	55186 → 443 [ACK] Seq=876 Ack=8809 Win=65280 Len=0
7090	551.87760	192.168.1.88	107.167.110.216	TCP	66	61411 → 443 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0
7123	552.47363	192.168.1.88	189.247.197.96	TCP	66	63638 → 443 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0
7288	651.47816	192.168.1.88	192.168.1.70	TCP	66	51359 → 443 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0
8334	896.48117	192.168.1.88	32.182.143.213	TCP	66	51350 → 443 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0
8455	969.48917	192.168.1.88	189.247.197.96	TCP	66	52875 → 443 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0

6455	319.289440900	2806:106e:2d:a245::c	2603:1030:40c:e::1	TCP	75	[TCP Keep-Alive] 49408 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
6456	319.347503200	2603:1030:40c:e::1	2806:106e:2d:a245::c	TCP	86	[TCP Keep-Alive ACK] 443 → 49408 [ACK] Seq=1 Ack=2 Win=8192 Len=0 SLE=1 SRE=2
6457	323.022641000	192.168.1.88	23.36.184.32	TCP	54	56487 → 443 [FIN, ACK] Seq=875 Ack=8809 Win=65280 Len=0
6458	323.030452600	23.36.184.32	192.168.1.88	TCP	54	[TCP Previous segment not captured] 443 → 56487 [FIN, ACK] Seq=8840 Ack=876 Win=64128 Len=0
6459	323.030452600	23.36.184.32	192.168.1.88	TCP	85	[TCP Retransmission] 443 → 56487 [PSH, ACK] Seq=8809 Ack=676 Win=64128 Len=31
6460	323.030516700	192.168.1.88	23.36.184.32	TCP	54	[TCP Dup ACK 5934#1] 56487 → 443 [ACK] Seq=876 Ack=8809 Win=65280 Len=0
6461	323.030601800	192.168.1.88	23.36.184.32	TCP	54	56487 → 443 [RST, ACK] Seq=876 Ack=8840 Win=0 Len=0
6466	326.645525300	2a06:98c1:3107:1681::	2806:106e:2d:a245::c	TLShv1.2	98	Application Data
6467	326.645934100	2806:106e:2d:a245::c	2a06:98c1:3107:1681::	TLShv1.2	102	Application Data
6468	326.652466800	2a06:98c1:3107:1681::	2806:106e:2d:a245::c	TCP	74	443 → 65127 [ACK] Seq=567 Ack=657 Win=17 Len=0
6471	335.651063100	2a06:98c1:3107:1681::	2806:106e:2d:a245::c	TLShv1.3	98	Application Data

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tcp.port == 443					
No.	Time	Source	Destination	Protocol	Length Info
3690	187.844357000	192.168.1.88	52.178.17.234	TLSv1.2	212 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
3691	187.989732000	52.178.17.234	192.168.1.88	TLSv1.2	105 Change Cipher Spec, Encrypted Handshake Message
3692	187.989732000	52.178.17.234	192.168.1.88	TLSv1.2	123 Application Data
3693	187.989917400	192.168.1.88	52.178.17.234	TCP	54 50196 → 443 [ACK] Seq=371 Ack=4356 Win=65280 Len=0
3694	187.992221900	192.168.1.88	52.178.17.234	TLSv1.2	141 Application Data
3695	187.992280300	192.168.1.88	52.178.17.234	TCP	1494 50196 → 443 [ACK] Seq=4394 Ack=3115 Win=4193536 Len=0
3696	187.992289300	192.168.1.88	52.178.17.234	TLSv1.2	162 Application Data
3697	187.992355500	192.168.1.88	52.178.17.234	TLSv1.2	92 Application Data
3698	187.992416900	192.168.1.88	52.178.17.234	TLSv1.2	1125 Application Data
3700	188.129800600	52.178.17.234	192.168.1.88	TLSv1.2	92 Application Data
3701	188.130159000	52.178.17.234	192.168.1.88	TCP	54 443 → 50196 [ACK] Seq=4394 Ack=2044 Win=4194816 Len=0
3702	188.130159000	52.178.17.234	192.168.1.88	TCP	54 443 → 50196 [ACK] Seq=4394 Ack=3115 Win=4193536 Len=0
3703	188.170775200	192.168.1.88	52.178.17.234	TCP	54 50196 → 443 [ACK] Seq=3115 Ack=4394 Win=65280 Len=0
3704	188.314103300	52.178.17.234	192.168.1.88	TLSv1.2	341 Application Data
3705	188.322156400	192.168.1.88	52.178.17.234	TCP	54 50196 → 443 [FIN, ACK] Seq=3115 Ack=4681 Win=65024 Len=0
3706	188.522205100	52.178.17.234	192.168.1.88	TCP	54 443 → 50196 [ACK] Seq=4681 Ack=3116 Win=4193536 Len=0
3707	188.522581300	52.178.17.234	192.168.1.88	TCP	54 [TCP Previous segment not captured] 443 → 50196 [FIN, ACK] Seq=4723 Ack=3116 Win=4193536 Len=0
3708	188.522581300	52.178.17.234	192.168.1.88	TCP	96 [TCP Retransmission] 443 → 50196 [PSH, ACK] Seq=4681 Ack=3116 Win=4193536 Len=42
3709	188.522643400	192.168.1.88	52.178.17.234	TCP	54 [TCP Dup ACK 3705#1] 50196 → 443 [ACK] Seq=3116 Ack=4681 Win=65024 Len=0
3710	188.522741200	192.168.1.88	52.178.17.234	TCP	54 50196 → 443 [RST, ACK] Seq=3116 Ack=4723 Win=0 Len=0
3721	197.772315800	2a06:98c1:3107::681	2806:106e:2d:a245::c	TLSv1.2	98 Application Data
3722	197.772615000	2a06:106e:2d:a245::c	2a06:98c1:3107::681	TLSv1.2	102 Application Data
3723	197.785018000	2a06:98c1:3107::681	2806:106e:2d:a245::c	TCP	74 443 → 60672 [ACK] Seq=1034 Ack=1181 Win=17 Len=0
3736	217.771159200	2a06:98c1:3107::681	2806:106e:2d:a245::c	TLSv1.2	98 Application Data
3737	217.771427800	2a06:106e:2d:a245::c	2a06:98c1:3107::681	TLSv1.2	102 Application Data
3738	217.785234600	2a06:98c1:3107::681	2806:106e:2d:a245::c	TCP	74 443 → 60672 [ACK] Seq=1058 Ack=1209 Win=17 Len=0
3739	222.728413300	2a06:106e:2d:a245::c	2600:1901:0:47fc::	TCP	75 [TCP Keep-Alive] 50506 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
3740	222.738408000	2600:1901:0:47fc::	2806:106e:2d:a245::c	TCP	86 [TCP Keep-Alive ACK] 443 → 50506 [ACK] Seq=1 Ack=2 Win=1045 Len=0 SLE=1 SRE=2
3742	230.156087300	192.168.1.88	107.167.96.39	TCP	55 [TCP Keep-Alive] 55946 → 443 [ACK] Seq=2300 Ack=5305 Win=64327 Len=1
3743	230.250208800	107.167.96.39	192.168.1.88	TCP	54 [TCP Keep-Alive ACK] 443 → 55946 [ACK] Seq=5205 Ack=2301 Win=60000 Len=0
<pre> Frame 518: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface Ubr Ethernet II, Src: zte 48:74:bc (d8:9a:0d:48:74:bc), Dst: Intel a4:2e:e1 (c4:0f:00:a4:2e:e1) Internet Protocol Version 4, Src: 23.36.184.32, Dst: 192.168.1.88 Transmission Control Protocol, Src Port: 443, Dst Port: 56620, Seq: 32, Ack: 2, Len: 0 </pre>					

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tcp.port == 443					
No.	Time	Source	Destination	Protocol	Length Info
496	33.149394100	192.168.1.88	23.36.186.55	TCP	54 56619 → 443 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
500	33.156271100	23.36.184.32	192.168.1.88	TCP	54 443 → 56631 [FIN, ACK] Seq=32 Ack=2 Win=497 Len=0
502	33.156271100	23.36.186.55	192.168.1.88	TCP	54 443 → 56619 [FIN, ACK] Seq=32 Ack=2 Win=501 Len=0
503	33.156271100	23.36.184.32	192.168.1.88	TCP	54 443 → 56601 [FIN, ACK] Seq=32 Ack=2 Win=497 Len=0
505	33.156271100	23.36.184.32	192.168.1.88	TLSv1.2	85 [TCP Retransmission], Encrypted Alert
506	33.156271100	23.36.186.55	192.168.1.88	TLSv1.2	85 [TCP Retransmission], Encrypted Alert
507	33.156271100	23.36.184.32	192.168.1.88	TLSv1.2	85 [TCP Retransmission], Encrypted Alert
509	33.156394700	192.168.1.88	23.36.184.32	TCP	54 [TCP Dup ACK 493#1] 56631 → 443 [ACK] Seq=2 Ack=1 Win=251 Len=0
510	33.156470600	192.168.1.88	23.36.186.55	TCP	54 [TCP Dup ACK 496#1] 56619 → 443 [ACK] Seq=2 Ack=1 Win=253 Len=0
511	33.156485900	192.168.1.88	23.36.184.32	TCP	54 [TCP Dup ACK 494#1] 56601 → 443 [ACK] Seq=2 Ack=1 Win=253 Len=0
512	33.156540100	192.168.1.88	23.36.184.32	TCP	54 56631 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
513	33.156577100	192.168.1.88	23.36.186.55	TCP	54 56619 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
514	33.156606400	192.168.1.88	23.36.184.32	TCP	54 56601 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
518	33.157781000	23.36.184.32	192.168.1.88	TCP	54 443 → 56620 [FIN, ACK] Seq=32 Ack=2 Win=497 Len=0
519	33.157781000	23.36.184.32	192.168.1.88	TLSv1.2	85 [TCP Retransmission], Encrypted Alert
520	33.157838400	192.168.1.88	23.36.184.32	TCP	54 [TCP Dup ACK 492#1] 56620 → 443 [ACK] Seq=2 Ack=1 Win=251 Len=0
521	33.157919200	192.168.1.88	23.36.184.32	TCP	54 56620 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
522	33.162560000	23.36.184.32	192.168.1.88	TCP	54 443 → 56620 [RST] Seq=1 Win=0 Len=0
523	33.165184900	23.36.184.32	192.168.1.88	TCP	54 443 → 56599 [FIN, ACK] Seq=32 Ack=2 Win=497 Len=0
524	33.165184900	23.36.184.32	192.168.1.88	TCP	54 443 → 56624 [FIN, ACK] Seq=32 Ack=2 Win=501 Len=0
525	33.165217300	192.168.1.88	23.36.184.32	TCP	54 [TCP Dup ACK 491#1] 56599 → 443 [ACK] Seq=2 Ack=1 Win=251 Len=0
526	33.165250900	192.168.1.88	23.36.184.32	TCP	54 [TCP Dup ACK 495#1] 56624 → 443 [ACK] Seq=2 Ack=1 Win=253 Len=0
527	33.165279600	23.36.184.32	192.168.1.88	TLSv1.2	85 [TCP Out-Of-Order], Encrypted Alert
528	33.165279600	23.36.184.32	192.168.1.88	TLSv1.2	85 [TCP Out-Of-Order], Encrypted Alert
529	33.165312200	192.168.1.88	23.36.184.32	TCP	54 56599 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
530	33.165355000	192.168.1.88	23.36.184.32	TCP	54 56624 → 443 [RST, ACK] Seq=2 Ack=32 Win=0 Len=0
557	37.770786700	2a06:98c1:3107::681	2806:106e:2d:a245::c	TLSv1.2	98 Application Data
558	37.771196200	2a06:106e:2d:a245::c	2a06:98c1:3107::681	TLSv1.2	102 Application Data
559	37.780800900	2a06:98c1:3107::681	2806:106e:2d:a245::c	TCP	74 443 → 60672 [ACK] Seq=526 Ack=597 Win=17 Len=0

No.	Time	Source	Destination	Protocol	Length	Info
45	2.620553600	192.168.1.88	20.96.153.111	TLSv1.2	529	Client Hello (SHA=fd.apl.iris.microsoft.com)
46	2.682640800	20.190.155.16	192.168.1.88	TCP	54	443 → 56643 [ACK] Seq=4006 Ack=3560 Win=4194816 Len=0
47	2.682648000	20.190.155.16	192.168.1.88	TCP	54	443 → 56643 [ACK] Seq=4006 Ack=7589 Win=4194816 Len=0
48	2.689982300	20.96.153.111	192.168.1.88	TCP	1506	443 → 56644 [ACK] Seq=1 Ack=476 Win=262144 Len=1452 [TCP PDU reassembled in 53]
49	2.689982300	20.96.153.111	192.168.1.88	TCP	1506	[TCP Previous segment not captured] 443 → 56644 [ACK] Seq=5809 Ack=476 Win=262144 Len=1452 [TCP PDU reassembled in 53]
50	2.689982300	20.96.153.111	192.168.1.88	TCP	1166	443 → 56644 [PSH, ACK] Seq=7261 Ack=476 Win=262144 Len=1112 [TCP PDU reassembled in 53]
51	2.689982300	20.96.153.111	192.168.1.88	TCP	1506	[TCP Retransmission] 443 → 56644 [ACK] Seq=1453 Ack=476 Win=262144 Len=1452 [TCP PDU reassembled in 53]
52	2.689982300	20.96.153.111	192.168.1.88	TCP	1506	[TCP Retransmission] 443 → 56644 [ACK] Seq=2809 Ack=476 Win=262144 Len=1452 [TCP PDU reassembled in 53]
53	2.689982300	20.96.153.111	192.168.1.88	TLSv1.2	1506	[TCP Retransmission], Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done
54	2.690054400	192.168.1.88	20.96.153.111	TCP	66	56644 → 443 [ACK] Seq=476 Ack=1453 Win=262144 Len=0 SLE=5809 SRE=7261
55	2.690098500	192.168.1.88	20.96.153.111	TCP	66	[TCP Dup ACK 54#1] 56644 → 443 [ACK] Seq=476 Ack=1453 Win=262144 Len=0 SLE=5809 SRE=8373
56	2.690112700	192.168.1.88	20.96.153.111	TCP	66	56644 → 443 [ACK] Seq=476 Ack=2905 Win=262144 Len=0 SLE=5809 SRE=8373
57	2.690122700	192.168.1.88	20.96.153.111	TCP	66	56644 → 443 [ACK] Seq=476 Ack=4357 Win=262144 Len=0 SLE=5809 SRE=8373
58	2.690138000	192.168.1.88	20.96.153.111	TCP	54	56644 → 443 [ACK] Seq=476 Ack=8373 Win=262144 Len=0
59	2.704195000	192.168.1.88	20.96.153.111	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
60	2.741544100	20.190.155.16	192.168.1.88	TCP	1506	443 → 56643 [ACK] Seq=4006 Ack=7589 Win=4194816 Len=1452 [TCP PDU reassembled in 62]
61	2.741544100	20.190.155.16	192.168.1.88	TCP	1506	443 → 56643 [ACK] Seq=5458 Ack=7589 Win=4194816 Len=1452 [TCP PDU reassembled in 62]
62	2.741544100	20.190.155.16	192.168.1.88	TLSv1.2	14574	Application Data
63	2.741622500	192.168.1.88	20.190.155.16	TCP	54	56643 → 443 [ACK] Seq=7589 Ack=21430 Win=65280 Len=0
64	2.772921600	20.96.153.111	192.168.1.88	TLSv1.2	185	Change Cipher Spec, Encrypted Handshake Message
65	2.772921600	20.96.153.111	192.168.1.88	TLSv1.2	123	Application Data
66	2.772983400	192.168.1.88	20.96.153.111	TCP	54	56644 → 443 [ACK] Seq=634 Ack=8493 Win=261888 Len=0
67	2.774717100	192.168.1.88	20.96.153.111	TLSv1.2	141	Application Data
68	2.774771500	192.168.1.88	20.96.153.111	TLSv1.2	92	Application Data
69	2.775954500	192.168.1.88	20.96.153.111	TCP	1494	56644 → 443 [ACK] Seq=759 Ack=8493 Win=261888 Len=1440 [TCP PDU reassembled in 71]
70	2.775954500	192.168.1.88	20.96.153.111	TCP	1494	56644 → 443 [ACK] Seq=2199 Ack=8493 Win=261888 Len=1440 [TCP PDU reassembled in 71]
71	2.775954500	192.168.1.88	20.96.153.111	TLSv1.2	311	Application Data
72	2.811025000	20.190.155.16	192.168.1.88	TCP	1506	443 → 56643 [ACK] Seq=21430 Ack=7589 Win=4194816 Len=1452 [TCP PDU reassembled in 75]
73	2.811025000	20.190.155.16	192.168.1.88	TCP	1506	443 → 56643 [ACK] Seq=22882 Ack=7589 Win=4194816 Len=1452 [TCP PDU reassembled in 75]
74	2.811171800	192.168.1.88	20.190.155.16	TCP	54	56643 → 443 [ACK] Seq=7589 Ack=74324 Win=65280 Len=0

```

Frame 4185: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 10
Ethernet II, Src: Intel_i422e1 (c4:0f:00:a4:2e:e1), Dst: zte_48:74:bc (d8:9a:0d:48:74:bc)
Internet Protocol Version 6, Src: 2086:106e:2d:a245:c0d2:83cc:644a:e7de, Dst: 2a06:98c1:3107:
Transmission Control Protocol, Src Port: 60672, Dst Port: 443, Seq: 1787, Ack: 1562, Len: 0
0000  d8 9a 0d 48 74 bc c4 0f 08 a4 2e e1 86 dd 00 02  ...HT.....E
0010  6c 64 00 14 06 40 28 06 10 6e 00 2d a2 45 c0 d2  ...ld...@...n...E...
0020  83 c0 64 4a e7 de 2a 06 98 c1 31 07 00 00 00 00  ...dJ...*...1.....
0030  00 00 68 12 27 15 ed 00 01 bb 8f e0 c0 fc d6 b7  ...h.....
0040  07 99 50 10 00 ff a2 46 00 00  ...P...F...

```

No.	Time	Source	Destination	Protocol	Length	Info
3831	280.546715300	192.168.1.162	192.168.1.254	ICMP	99	Destination unreachable (Port unreachable)
3832	286.141080500	192.168.1.162	192.168.1.254	DNS	71	Standard query 0xd092 A 192.168.1.X
3835	291.142072100	fe80::3a48:2b54:ea4...	fe80::1	DNS	91	Standard query 0xd092 A 192.168.1.X
3836	291.546903400	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0xd092 A 192.168.1.X
3844	296.141540600	192.168.1.162	192.168.1.254	DNS	71	Standard query 0xd092 A 192.168.1.X
3854	301.547053400	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0xd092 Refused A 192.168.1.X
3855	301.547112100	192.168.1.162	192.168.1.254	ICMP	99	Destination unreachable (Port unreachable)
3856	305.164640000	192.168.1.162	192.168.1.254	DNS	71	Standard query 0x38ee A 192.168.1.X
3859	310.207608400	fe80::3a48:2b54:ea4...	fe80::1	DNS	91	Standard query 0x38ee A 192.168.1.X
3860	310.547157300	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0x38ee Refused A 192.168.1.X
3865	315.256417600	192.168.1.162	192.168.1.254	DNS	71	Standard query 0x38ee A 192.168.1.X
3866	315.257265100	fe80::3a48:2b54:ea4...	fe80::1	DNS	91	Standard query 0x38ee A 192.168.1.X
3874	320.547795800	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0x38ee Refused A 192.168.1.X
3875	320.548881700	192.168.1.162	192.168.1.254	ICMP	99	Destination unreachable (Port unreachable)
3876	325.266895300	192.168.1.162	192.168.1.254	DNS	71	Standard query 0xb976 A 192.168.1.X
3879	330.287651200	fe80::3a48:2b54:ea4...	fe80::1	DNS	91	Standard query 0xb976 A 192.168.1.X
3882	330.547868000	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0xb976 A 192.168.1.X
3885	335.297643300	192.168.1.162	192.168.1.254	DNS	71	Standard query 0xb976 A 192.168.1.X
3889	340.560646400	192.168.1.162	192.168.1.254	DNS	71	Standard query response 0xb976 Refused A 192.168.1.X
3890	340.561704500	192.168.1.162	192.168.1.254	ICMP	99	Destination unreachable (Port unreachable)
3891	345.403768100	192.168.1.162	192.168.1.254	DNS	71	Standard query 0xf34f A 192.168.1.X
3894	350.476119000	fe80::3a48:2b54:ea4...	fe80::1	DNS	91	Standard query 0xf34f A 192.168.1.X
3895	350.550438200	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0xf34f A 192.168.1.X
3896	355.407624000	192.168.1.162	192.168.1.254	DNS	71	Standard query 0xf34f A 192.168.1.X
3900	360.522874400	192.168.1.162	192.168.1.254	DNS	71	Standard query 0x23e5 A 192.168.1.X
3909	360.547878700	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0xf34f Refused A 192.168.1.X
3910	360.548324000	192.168.1.162	192.168.1.254	ICMP	99	Destination unreachable (Port unreachable)
3913	365.534918400	fe80::3a48:2b54:ea4...	fe80::1	DNS	91	Standard query 0x23e5 A 192.168.1.X
3914	365.548513200	192.168.1.254	192.168.1.162	DNS	71	Standard query response 0x23e5 Refused A 192.168.1.X
3915	370.555009900	192.168.1.162	192.168.1.254	DNS	71	Standard query 0x23e5 A 192.168.1.X

```

Frame 3678: Packet, 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface 10
Ethernet II, Src: Intel_i422e1 (c4:0f:00:a4:2e:e1), Dst: zte_48:74:bc (d8:9a:0d:48:74:bc)
Internet Protocol Version 4, Src: 192.168.1.162, Dst: 192.168.1.254
Internet Control Message Protocol
0000  d8 9a 0d 48 74 bc c4 0f 08 a4 2e e1 00 00 45 c0  ...HT.....E
0010  00 55 0d 6b 00 00 40 01 e7 8c c0 a8 01 a2 c0 a8  ...U...k...@.....
0020  01 fe 03 03 82 24 00 00 00 00 45 00 00 39 6e b9  ......$.....E...9n
0030  40 00 40 11 47 0a c0 a8 01 fe c0 a8 01 a2 00 35  ...@...G.....5
0040  a9 34 00 25 17 15 be 58 81 85 00 01 00 00 00 00  ...4...%...X.....
0050  00 00 03 21 39 32 03 31 36 38 01 31 01 58 00 00  ......192.168.1.X...
0060  01 00 01  ...

```

10. Conclusión

Se identificó actividad consistente con **canal C2 encubierto en tráfico HTTPS**, precedido por reconocimiento.

El patrón observado indica un riesgo elevado que requiere **contención y análisis adicional del host** para descartar compromiso activo.